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The Editors
Nature Machine Intelligence

Dear Editors,

We are pleased to submit our manuscript, “**Useful possibility collapse: an intervention-aware framework for adaptive emergence in learning systems**”, for consideration as an Article in *Nature Machine Intelligence*.

“Emergence” is measured differently in complex-systems science (causal efficacy of macro-variables), information theory (synergy), and machine learning (abrupt ability curves)—and the communities disagree about what counts. The dispute is not academic: whether large models have “emergent abilities” is currently argued by reading performance curves, although a curve alone carries no information about where an ability came from.

Our manuscript addresses this gap in three steps.

First, an honest theory architecture. The trajectory-space quantity we start from is a known specific-information object (the Bayesian-surprise divergence), and we position it as such: it fixes what emergence claims must be about—distributions over futures—but does not by itself carry a verdict. The verdict comes from two layered criteria stated on a declared observer contract: *structural collapse* (potential, conditional selectivity, specificity) and *adaptive qualification* (usefulness, endogeneity, acquisition). The layering is testable and tested: under a second plausible observer contract, all 60 control verdicts and 14/15 structural verdicts are unchanged, while value verdicts flip conservatively in five cases, exactly the contract-relativity the definition declares. The signatures audited here—performance jumps, representation jumps, PID-style synergy, Rosas’ Ψ , and Hoel’s effective information—are lossy projections of the same trajectory-law family; we compute the latter two from their published equations with zero Monte-Carlo error on fully enumerated Markov chains, within declared candidate families, and we do not claim these formalisms are invalid in their home use cases.

Second, comparisons with matched degrees of freedom. Beyond single-signal audits, we give the prior signatures equal or greater freedom—multi-threshold conjunctions, logistic regression and decision trees over all signals including the exact EI and Ψ —and test frozen transfer to fresh systems. Every fitted baseline caps at 0.9 and misclassifies the true conditional emergence, while the frozen six-component protocol scores 10/10 on the same fresh seed. Two independent domains also exposed the same methodological principle, which we believe will matter for capability auditing: in converged systems, the behaving policy prices its own triggers into the baseline, so useful emergence is visible through counterfactual contrast rather than observed improvement along the behaving trajectory.

Third, stress tests across complementary evidence types. Under internally frozen author-maintained protocols (with all thirty-one recorded misses retained), the full six-component protocol is tested in three built domains and one public benchmark. It classifies 25/25 fresh external-swarm verdicts; in Contextual Level-Based Foraging, 9/10 learned policies pass while all 40 initialization/scripted controls reject; and in a controlled latent-context sequence task, 10/10 next-token-trained transformers pass while all 40 controls reject. Decisively, in *unmodified* Overcooked-AI—a public third-party coordination benchmark—the complete protocol, thresholds and five predictions were frozen with a *public external timestamp* that predates every confirmatory measurement, and all five registered predictions

passed: learned self-play policies are accepted on exactly the registered 8/12 seed line, all 48 control verdicts are rejections with the failure routes the layered definition predicts, and the primary seed-level usefulness do-contrast is positive on 12/12 seeds ($p = 2 \times 10^{-4}$). The retained misses are used to delimit scope rather than repaired. For example, persistence is stable under declared temporal and observational perturbations but not under novel geometries, and the sequence model generalizes to distractors but not to unseen marker positions. Separately, a frozen future-distribution score prospectively discovers value-critical decisions in uncurated chess positions across two months (AUROC 0.730 and 0.725), transfers to a classical-versus-NNUE referee, and is directionally consistent under an engine-free realized-outcome referee whose registered interaction we report as an underpowered null. The process proxy recovers grokking and induction-head transitions and transfers through Pythia-1.4B, while its registered 2.8B and thinning failures expose checkpoint-resolution limits.

We believe the work fits *Nature Machine Intelligence*’s scope directly: it turns “did an ability emerge?” from a curve-reading exercise into an intervention-aware measurement programme, with relevance to capability auditing and forecasting in machine learning, and with a transparency standard (frozen thresholds, registered failures, full code and outputs) that we hope raises the bar for definitional claims in this area.

The manuscript is not under consideration elsewhere. All code, outputs, prospectively frozen protocols, a reproducibility manifest, invalid-data registry, and a complete prediction ledger sufficient to audit every figure accompany the submission.

Suggested referees: [names spanning the three communities—causal emergence, information decomposition, and LLM capability evaluation—to be filled in].

Sincerely,

[Corresponding author]
on behalf of all authors